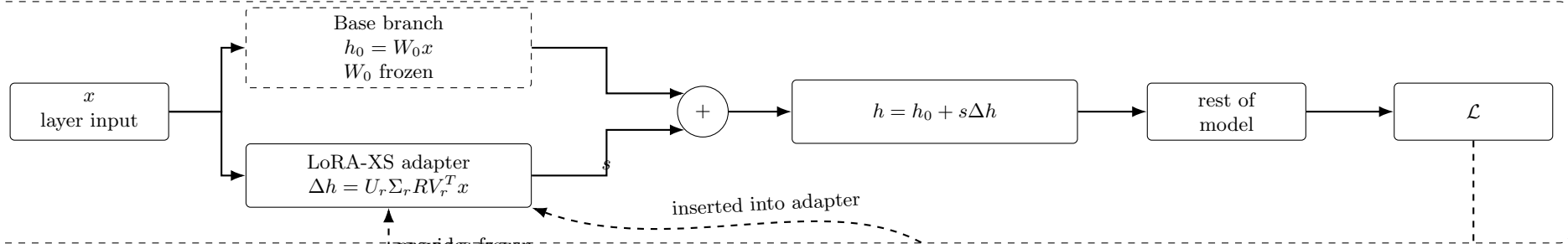


LoRA-XS Computation Graph

Forward pass

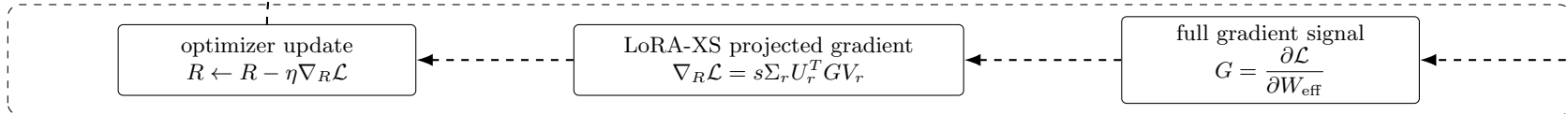


provides frozen
 U_r, Σ_r, V_r

Pre-computed frozen SVD basis
from pretrained weight W_0
 $W_0 \approx U_r \Sigma_r V_r^T$

only trainable
matrix R

Backward pass



$$W_{\text{eff}} = W_0 + s U_r \Sigma_r R V_r^T$$